

Lab Template Stat 240

Lab 02

AUTHOR

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Problem 1

Part a)

```
median_func <- function(vector){  
  
  sorted <- sort(vector)  
  len <- length(sorted)  
  if(len %% 2 == 0){  
    num1 <- sorted[(len/ 2)+1]  
    num2 <- sorted[len / 2 ]  
    return ((num1+num2) / 2)  
  
  }  
  {  
    val <- sorted[(len/2)+1]  
    return (val)  
  }  
  
}  
  
median_func(c(1,2,3,4, 5))
```

```
[1] 3
```

Part b)

```
## write answer here  
a <- (301611434 %% 27) + 1  
library(readr)  
library(tidyverse)
```

— Attaching core tidyverse packages — tidyverse 2.0.0 —

✓ dplyr	1.1.4	✓ purrr	1.2.1
✓ forcats	1.0.1	✓ stringr	1.6.0
✓ ggplot2	4.0.1	✓ tibble	3.3.1
✓ lubridate	1.9.4	✓ tidyr	1.3.2

— Conflicts — tidyverse_conflicts() —

- ✖ dplyr::filter() masks stats::filter()
- ✖ dplyr::lag() masks stats::lag()

i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

```
data <- read_tsv("AS-N100.tsv")
```

Rows: 10000 Columns: 10

— Column specification —

Delimiter: "\t"

chr (1): ticker

dbl (7): open, high, low, close, volume, quantity, tradecount

date (1): date

time (1): timestamp

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
symbols <- sort(unique(data$ticker))[a]  
  
worked <- filter(data, ticker == symbols)  
count <- c(worked$open)  
  
median_func(count)
```

[1] 106.99